

VOLUME 3

SOLUTION 0-2

Cicada 3301 - Liber Primus pages 0-2

Geometric mirror navigation / continuation research



27 x 27 matrix / mirrored structures / Euler totient / Mobius phase / GP mod 29

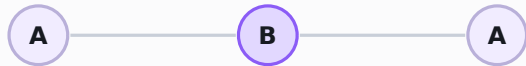
SECTION 01

A geometric direction rule

I'm currently working on Volume 3, and one of the main ideas I'm testing is that the maze may follow a very simple geometric rule

Cicada may have used mirrored structures to determine direction automatically.

For a horizontal mirror:



- land on the left A → move right, toward the center
- land on the right A → move left, toward the center

For a vertical mirror:



- land on the top A → move down, toward the center
- land on the bottom A → move up, toward the center

CORE RULE

So if you land on an outer rune, the mirror itself tells you which way to move: toward the center.

If you land on the center, the mirror cannot choose a direction because both sides are equal. In that case, its totient signature may instead tell you which structure to find next.

Example 1

EA – A – EA → 12 – 8 – 12
 NG – T – NG → 12 – 8 – 12

The same totient signature links the two structures.

Example 2

I – NG – I → 4 – 12 – 4
 H – NG – C → 4 – 12 – 4

SECTION 02

How the rule behaves

Again, the hidden key matches the signature of another structure.

The core idea is simple: if the route lands on an outer rune, the active key is a 3-rune node. If it lands exactly on the center, a hidden-key search is triggered instead.

GEOMETRIC DECISION

If we land on an outer rune of a node, we simply follow the node toward its center — the direction is already defined by the geometry.

During a hidden-key search, the movement always brings us to the center of the key. From there, the key itself appears on the perpendicular axis, so we turn 90° to find its two outer runes.

This seems to be how the algorithm works, and it is also the simplest geometric rule.

The mirrors may have been chosen not just for symmetry, but because they naturally define direction.

It is simple, elegant, and so far it works.

Structural observations

Today I looked more closely at the 27×27 matrix itself. It contains a lot of interesting connections, but most of them seem to confirm what we already found rather than reveal anything new.

One unusual thing is that the matrix contains exactly 77 mirrored structures of the form A-B-A. That surprised me, because we have already seen several connections involving the number 7. It is possible that 77 was intentional.

In Western esoteric and occult symbolism, the number 12 is often associated with order, completeness, and the organization of a whole system.

12 signs of the zodiac, 12 months, 12 tribes of Israel, 12 apostles, in some mystical systems, twelve represents the division of unity into a complete and ordered structure

That makes the central structure

P -12- NG -12- P

especially interesting symbolically

if the connection to the number 12 is intentional, it could indirectly support the validity of our result

BREAKTHROUGH

I think I found an extremely strong continuation, and our theory is correct

SECTION 03

CRY endpoint and structural value 6

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27									
1	S/Z	H	E	O	G	M	IA/IO	F	1	S/Z	Y	E	NG	C/K	13	TH	J	G	AE	1	F	J	OE	1	O	N	TH	1	N	E	W	B				
2	A	S/Z	OE	EO	I	NG	U/V	1	N	AE	G	I	TH	1	H	B	N	IA/IO	1	P	B	N	1	EO	J	TH	1	A	NG	Y	X	1	B			
3	D	P	1	E	IA/IO	F	X	G	F	1	S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/IO	1	Y	H	1	B	U/V	1	G	O	P	1	NG			
4	AE	TH	1	I	AE	H	X	TH	P	G	1	Y	F	NG	C/K	J	1	TH	A	M	I	NG	I	IA/IO	Y	1	H	EO	1	T	H	AE				
5	IA/IO	1	X	Y	1	E	U/V	NG	1	C/K	F	N	1	EO	TH	J	I	TH	1	TH	P	1	Y	NG	EA	1	Y	N	E	Y	1	D	X	NG	W	1
6	B	C/K	X	D	B	F	1	M	T	N	E	1	EA	J	N	L	G	B	1	X	G	TH	4	Y	I	D	A	1	NG	G	M	J				
7	R	O	L	EO	1	Y	T	P	I	TH	D	J	1	OE	H	L	1	EA	1	F	X	AE	P	G	X	1	F	L	W	EO	T	AE	1			
8	F	G	N	1	M	EO	L	N	1	NG	M	F	R	IA/IO	1	A	S/Z	1	EA	M	NG	X	X	EO	D	B	1	OE	D	M	O	1	EA			
9	EO	H	1	D	NG	G	1	OE	NG	L	U/V	R	N	T	1	AE	S/Z	Y	U/V	H	R	T	1	H	C/K	1	S/Z	OE	T	OE	NG	U/V				
10	R	1	S/Z	1	F	O	E	W	EA	OE	L	F	AE	1	M	R	NG	1	D	A	M	R	1	W	A	I	M	S/Z	N	1	S/Z	OE	R			
11	U/V	W	S/Z	L	IA/IO	4	OE	A	AE	NG	S/Z	D	P	T	1	C/K	R	TH	IA/IO	1	R	B	O	D	TH	F	1	Y	X	I	T	4				
12	OE	I	1	F	L	O	1	F	EA	1	R	O	OE	M	H	M	G	P	H	1	AE	H	N	O	H	Y	1	OE	S/Z	L	1	N	G	1		
13	AE	Y	1	OE	C/K	B	L	J	1	NG	L	OE	1	AE	J	EA	W	1	EA	TH	O	1	B	OE	Y	1	C/K	F	C/K	J	1	L	AE	4	N	1
14	TH	P	1	U/V	X	1	OE	X	G	1	P	F	S/Z	EO	AE	OE	1	NG	P	EO	O	E	A	1	G	AE	IA/IO	NG	TH	O	1	P	A			
15	OE	TH	R	NG	AE	1	C/K	S/Z	B	EO	Y	H	1	T	X	E	L	R	4	G	W	Y	1	J	F	I	N	IA/IO	C/K	Y	EA					
16	I	IA/IO	1	O	TH	1	E	C/K	AE	C/K	X	IA/IO	EA	1	O	L	C/K	1	F	R	D	NG	E	U/V	1	D	C/K	L	EA	S/Z	X	C/K				
17	G	IA/IO	4	W	S/Z	TH	1	F	D	NG	1	I	IA/IO	M	A	AE	G	1	W	S/Z	1	N	D	1	C/K	P	TH	X	P	EA	EA	1	W	U/V	1	
18	EA	W	1	F	W	J	Y	1	X	D	W	C/K	G	C/K	OE	1	D	X	OE	1	R	IA/IO	G	1	N	P	A	Y	P	1	C/K	Y	H	1		
19	F	E	J	EA	N	1	OE	AE	1	U/V	A	1	H	R	1	E	EA	TH	F	J	A	1	L	X	S/Z	T	1	M	F	L	F	T	1	G		
20	I	TH	1	U/V	L	G	1	X	EA	T	S/Z	L	J	P	1	L	X	I	OE	M	1	U/V	4	H	T	1	B	EO	L	D	H	B	M	1		
21	EA	R	B	1	N	H	B	E	G	EO	1	D	L	W	EO	IA/IO	P	O	1	H	E	EA	1	W	I	R	I	H	1	U/V	TH	H				
22	Y	1	N	X	B	G	J	P	U/V	1	NG	EA	F	N	I	E	D	IA/IO	NG	R	1	D	B	J	IA/IO	OE	M	I	1	F	T	1				
23	J	D	I	TH	R	L	S/Z	1	E	EO	O	G	B	T	D	1	TH	A	N	C/K	Y	1	IA/IO	S/Z	TH	D	1	NG	F	L	E					
24	G	H	C/K	1	E	O	I	T	N	X	1	P	T	F	H	R	D	E	F	T	1	AE	W	H	1	B	C/K	1	F	1	P	A	L			
25	U/V	EA	N	L	J	1	J	C/K	L	W	EA	D	U/V	D	EO	1	EA	X	D	W	H	EA	1	TH	IA/IO	AE	C/K	L	T	W						
26	E	I	C/K	1	P	OE	D	C/K	1	N	H	A	1	R	I	G	TH	EA	E	T	G	1	TH	H	O	IA/IO	W	AE	J	E	1	NG				
27	EA	D	1	O	AE	1	A	L	1	B	J	C/K	U/V	X	T	A	B	J	P	1	F	EA	1	H	D	N	IA/IO	U/V	P	S/Z	U/V	W	13			

Matrix state at the CRY endpoint and the structures active around the central route.

After decrypting CRY, the reading ends at S(17,16).

The value 6 is not introduced at this point arbitrarily. It comes directly from the structure that produced CRY: the preceding mirror E-X-E is transformed into E-G-E, whose totient signature is 6-2-6. The outer value 6 is then used as the movement distance that leads to the CRY-reading node.

Therefore, when CRY ends at S(17,16), 6 is the immediately preceding structural movement value. The next question is whether the new endpoint itself confirms that same value geometrically.

ANSWER

It does.

SECTION 04

The radius-6 mirror confirms itself

S(17,16) is one endpoint of an exact radius-6 mirror:

$$S(17,4) \text{ --6-- } IA/O(17,10) \text{ --6-- } S(17,16)$$

So the same value 6 that was produced by the previous totient structure reappears locally at the endpoint as an exact geometric radius. In other words, we do not simply decide to move six cells from S(17,16): the position of S(17,16) itself identifies a radius-6 structure.

The center of this mirror, IA/O(17,10), independently confirms the same number through the totient operation:

$$\begin{aligned} IA/O &= 27 \\ \phi(27) &= 18 \\ \phi(18) &= 6 \end{aligned}$$

Therefore:

$$\phi^2(IA/O) = 6$$

This means that the radius of the mirror is numerically encoded by its own central rune.

The same center IA/O(17,10) then participates in another exact mirror of radius 6:

$$TH(11,16) \text{ --6-- } IA/O(17,10) \text{ --6-- } TH(23,4)$$

We therefore have two different radius-6 mirrors sharing exactly the same center:

$$\begin{aligned} S(17,4) \text{ --6-- } IA/O(17,10) \text{ --6-- } S(17,16) \\ TH(11,16) \text{ --6-- } IA/O(17,10) \text{ --6-- } TH(23,4) \end{aligned}$$

This suggests a natural transition: once the path reaches S(17,16) on the first radius-6 mirror, it can transfer through the shared IA/O center to the companion radius-6 mirror.

SECTION 05

A forced transition to TH(11,16)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	
1	S/Z	H	E	O	G	M	IA/O	F	1 S/Z	Y	E	NG	C/K	13 TH	J	G	AE	1 F	J	OE	1 O	N	TH	1 N	E	W	B	
2	A	S/Z	OE	EO	I	NG	U/V	1 N	AE	G	I	TH	1 H	B	N	IA/O	P	B	N	1 EO	J	TH	1 A	NG	Y	X	1 B	
3	D	P	1 E	IA/O	F	X	G	F	1 S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/O	Y	H	1 B	U/V	1 G	O	P	1 NG	
4	AE	TH	1 I	AE	H	X	TH	P	G	1 Y	F	NG	C/K	J	1 TH	A	M	I	NG	I	IA/O	Y	1 H	EO	1 T	H	AE	
5	IA/O	1 X	Y	1 E	U/V	NG	1 C/K	F	N	1 EO	TH	J	I	TH	1 TH	P	1 Y	NG	EA	1 Y	N	E	Y	1 D	X	NG	W	1
6	B	C/K	X	D	B	F	1 M	T	N	E	1 EA	J	N	L	G	B	1 X	G	TH	4 Y	I	D	A	1 NG	G	M	J	
7	R	O	L	EO	1 Y	T	P	I	TH	D	J	1 OE	H	L	1 EA	1 F	X	AE	P	G	X	1 F	L	W	EO	T	AE	1
8	F	G	N	1 M	EO	L	N	1 NG	M	F	R	IA/O	1 A	S/Z	1 EA	M	NG	X	X	EO	D	B	1 OE	D	M	O	1 EA	
9	EO	H	1 D	NG	G	1 OE	NG	L	U/V	R	N	T	1 AE	S/Z	Y	U/V	H	R	T	1 H	C/K	1 S/Z	OE	T	OE	NG	U/V	
10	R	1 S/Z	1 F	O	E	W	EA	OE	L	F	AE	1 M	R	NG	1 D	A	M	R	1 W	A	I	M	S/Z	N	1 S/Z	OE	R	
11	U/V	W	S/Z	L	IA/O	4 OE	A	AE	NG	S/Z	D	P	T	1 C/K	R	TH	IA/O	R	B	O	D	TH	F	1 Y	X	I	T	4
12	OE	I	1 F	L	O	1 F	EA	1 R	O	OE	M	H	M	G	P	H	1 AE	H	N	O	H	Y	1 OE	S/Z	L	1 N	G	1
13	AE	Y	1 OE	C/K	B	L	J	1 NG	L	OE	AE	J	EA	W	1 EA	TH	O	1 B	OE	Y	1 C/K	F	C/K	J	1 L	AE	4 N	1
14	TH	P	1 U/V	X	1 OE	X	G	1 P	F	S/Z	EO	AE	OE	1 NG	P	EO	O	E	A	1 G	AE	IA/O	NG	TH	O	1 P	A	
15	OE	TH	R	NG	AE	1 C/K	S/Z	B	EO	Y	H	1 T	X	E	L	R	4 G	W	Y	1 J	F	I	N	IA/O	C/K	Y	EA	
16	I	IA/O	1 O	TH	1 E	C/K	AE	C/K	X	IA/O	EA	1 O	L	C/K	1 F	R	D	NG	E	U/V	1 D	C/K	L	EA	S/Z	X	C/K	
17	G	IA/O	4 W	S/Z	TH	1 F	D	NG	1 I	IA/O	M	A	AE	G	1 W	S/Z	1 N	D	1 C/K	P	TH	X	P	EA	EA	1 W	U/V	1
18	EA	W	1 F	W	J	Y	1 X	D	W	C/K	G	C/K	OE	1 D	X	OE	1 R	IA/O	G	1 N	P	A	Y	P	1 C/K	Y	H	1
19	F	E	J	EA	N	1 OE	AE	1 U/V	A	1 H	R	1 E	EA	TH	F	J	A	1 L	X	S/Z	T	1 M	F	L	F	T	1 G	
20	I	TH	1 U/V	L	G	1 X	EA	T	S/Z	L	J	P	1 L	X	I	1 OE	M	1 U/V	4 H	T	1 B	EO	L	D	H	B	M	1
21	EA	R	B	1 N	H	B	E	G	EO	1 D	L	W	EO	IA/O	P	O	1 H	E	EA	1 W	I	R	I	H	1 U/V	TH	H	
22	Y	1 N	X	B	G	J	P	U/V	1 NG	EA	F	N	I	E	D	IA/O	NG	R	1 D	B	J	IA/O	OE	M	I	1 F	T	1
23	J	D	I	TH	1 R	L	S/Z	1 E	EO	O	G	B	T	D	1 TH	A	N	C/K	Y	1 IA/O	S/Z	TH	D	1 NG	F	L	E	
24	G	H	C/K	1 E	O	I	T	N	X	1 P	T	F	H	R	D	E	F	T	1 AE	W	H	1 B	C/K	1 F	1 P	A	L	
25	U/V	EA	N	L	J	1 J	C/K	L	W	EA	D	U/V	D	EO	1 EA	X	D	W	H	EA	1 TH	IA/O	AE	C/K	L	T	W	
26	E	I	C/K	1 P	OE	D	C/K	1 N	H	A	1 R	I	G	TH	EA	E	T	G	1 TH	H	O	IA/O	W	AE	J	E	1 NG	
27	EA	D	1 O	AE	1 A	L	1 B	J	C/K	U/V	X	T	A	B	J	P	1 F	EA	1 H	D	N	IA/O	U/V	P	S/Z	U/V	W	13

Updated matrix state: purple plaintext route, blue hidden-key clues, and green geometric confirmation points.

Now the direction is no longer arbitrary. From the current point S(17,16), only one endpoint of the companion mirror can be reached by a single straight movement of exactly 6 cells:

S(17,16) → U6 → TH(11,16)

CONTINUATION

And this is exactly the point from which I found the real continuation — the Möbius phase matched, and everything else matched as well

Once the transition reaches TH(11,16), we read diagonally down-right along the axis of the A-TH-A mirror.

The five ciphertext runes are:

TH(11,16) → AE(12,17) → B(13,18) → A(14,19) → J(15,20)

So the ciphertext sequence is:

TH – AE – B – A – J

SECTION 06

Key generation and Möbius phase

The key has already been generated from the preceding OE-J-OE structure:

OE-J-OE $\rightarrow$ OE- $\varphi(J)$ -OE
 Since $J = 11$ and $\varphi(11) = 10 = I$, this gives:
 OE-I-OE

Its Möbius phase is then calculated from the totient values:

$$\mu(10) + \mu(4) + \mu(10) = 1 + 0 + 1 = 2 \bmod 3$$

so the required phase is phase 2.

For the key OE-I-OE, phase 2 gives:

OE-OE-I

Repeating this key across the five ciphertext runes gives:

OE – OE – I – OE – OE

Now decrypt using the standard Gematria Primus subtraction:

$$P = C - K \bmod 29$$

SECTION 07

Decrypting NOW THE

Rune by rune:

TH(2) - OE(22) = 9 = N
AE(25) - OE(22) = 3 = O
B(17) - I(10) = 7 = W
A(24) - OE(22) = 2 = TH
J(11) - OE(22) = 18 = E

NOW THE

So, after the geometric transition lands on TH(11,16), the already-determined Möbius phase fixes the key before the plaintext is inspected, and the diagonal beginning at that exact point decrypts directly to NOW THE.

The important point is that the whole process is sequential:

previous geometric structure → transformed key structure → Möbius phase → next
geometric reading → plaintext

SECTION 08

Phrase-level continuity

“AS I GO, THE WEATHER TURNS COLD. I MAY CRY NOW. THE ...”

Again, I noticed the same subtle pattern, phrases containing “the” seem to be generated together with it, like AS I GO THE and now NOW THE

So semantically there may be a comma or full stop between them, but the method already starts generating the beginning of the next block

What surprises me even more is how each structure keeps leading into the center of yet another new mirror structure



SECTION 09

Visual notation from this point onward

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1	S/Z	H	E	O	G	M	IA/IO	F	S/Z	Y	E	NG	C/K	TH	J	G	AE	F	J	OE	O	N	TH	N	E	W	B
2	A	S/Z	OE	EO	I	NG	U/V	N	AE	G	I	TH	H	B	N	IA/IO	P	B	N	EO	J	TH	A	NG	Y	X	B
3	D	P	E	IA/IO	F	X	G	F	S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/IO	Y	H	B	U/V	G	O	P	NG
4	AE	TH	I	AE	H	X	TH	P	G	Y	F	NG	C/K	J	TH	A	M	I	NG	I	IA/IO	Y	H	EO	T	H	AE
5	IA/IO	X	Y	E	U/V	NG	C/K	F	N	EO	TH	J	I	TH	TH	P	Y	NG	EA	Y	N	E	Y	D	X	NG	W
6	B	C/K	X	D	B	F	M	T	N	E	EA	J	N	L	G	B	X	G	TH	Y	I	D	A	NG	G	M	J
7	R	O	L	EO	Y	T	P	I	TH	D	J	OE	H	L	EA	F	X	AE	P	G	X	F	L	W	EO	T	AE
8	F	G	N	M	EO	L	N	NG	M	F	R	IA/IO	A	S/Z	EA	M	NG	X	X	EO	D	B	OE	D	M	O	EA
9	EO	H	D	NG	G	OE	NG	L	U/V	R	N	T	AE	S/Z	Y	U/V	H	R	T	H	C/K	S/Z	OE	T	OE	NG	U/V
10	R	S/Z	F	O	E	W	EA	OE	L	F	AE	M	R	NG	D	A	M	R	W	A	I	M	S/Z	N	S/Z	OE	R
11	U/V	W	S/Z	L	IA/IO	OE	A	AE	NG	S/Z	D	P	T	C/K	R	TH	IA/IO	R	B	O	D	TH	F	Y	X	I	T
12	OE	I	F	L	O	F	EA	R	O	OE	M	H	M	G	P	H	AE	H	N	O	H	Y	OE	S/Z	L	N	G
13	AE	Y	OE	C/K	B	L	J	NG	L	OE	AE	J	EA	W	EA	TH	O	B	OE	Y	C/K	F	C/K	J	L	AE	N
14	TH	P	U/V	X	OE	X	G	P	F	S/Z	EO	AE	OE	NG	P	EO	O	E	A	G	AE	IA/IO	NG	TH	O	P	A
15	OE	TH	R	NG	AE	C/K	S/Z	B	EO	Y	H	T	X	E	L	R	G	W	Y	J	F	I	N	IA/IO	C/K	Y	EA
16	I	IA/IO	O	TH	E	C/K	AE	C/K	X	IA/IO	EA	O	L	C/K	F	R	D	NG	E	U/V	D	C/K	L	EA	S/Z	X	C/K
17	G	IA/IO	W	S/Z	TH	F	D	NG	I	IA/IO	M	A	AE	G	W	S/Z	N	D	C/K	P	TH	X	P	EA	EA	W	U/V
18	EA	W	F	W	J	Y	X	D	W	C/K	G	C/K	OE	D	X	OE	R	IA/IO	G	N	P	A	Y	P	C/K	Y	H
19	F	E	J	EA	N	OE	AE	U/V	A	H	R	E	EA	TH	F	J	A	L	X	S/Z	T	M	F	L	F	T	G
20	I	TH	U/V	L	G	X	EA	T	S/Z	L	J	P	L	X	I	OE	M	U/V	H	T	B	EO	L	D	H	B	M
21	EA	R	B	N	H	B	E	G	EO	D	L	W	EO	IA/IO	P	O	H	E	EA	W	I	R	I	H	U/V	TH	H
22	Y	N	X	B	G	J	P	U/V	NG	EA	F	N	I	E	D	IA/IO	NG	R	D	B	J	IA/IO	OE	M	I	F	T
23	J	D	I	TH	R	L	S/Z	E	EO	O	G	B	T	D	TH	A	N	C/K	Y	IA/IO	S/Z	TH	D	NG	F	L	E
24	G	H	C/K	E	O	I	T	N	X	P	T	F	H	R	D	E	F	T	AE	W	H	B	C/K	F	P	A	L
25	U/V	EA	N	L	J	J	C/K	L	W	EA	D	U/V	D	EO	EA	X	D	W	H	EA	TH	IA/IO	AE	C/K	L	T	W
26	E	I	C/K	P	OE	D	C/K	N	H	A	R	I	G	TH	EA	E	T	G	TH	H	O	IA/IO	W	AE	J	E	NG
27	EA	D	O	AE	A	L	B	J	C/K	U/V	X	T	A	B	J	P	F	EA	H	D	N	IA/IO	U/V	P	S/Z	U/V	W

The updated matrix with the three-color notation and thin purple mirror borders.

For clarity, from now on the colors will mean the following:

PURPLE

ciphertext that turns into actual plaintext.

BLUE

hidden key points, used for their geometry as a clue.

GREEN

structures geometrically connected to a point or pointing toward something.

Such green points may also serve as potential clues in the future.

There will be only a few green points. I won't mark everything, only the most unique ones that genuinely helped in finding or confirming the plaintext.

I'll also use a thin purple border only for mirrors directly involved in forming or confirming the plaintext, since they may serve as potential return points, as with H TH H (I MAY branch).

SECTION 10

A literary observation

What is especially interesting here is the way the external world seems to reflect the speaker's inner state. In literary terms, this resembles what is sometimes called pathetic fallacy, where nature or weather mirrors a character's emotions.

**AS I GO, THE WEATHER TURNS COLD.
I MAY CRY NOW. THE ...**